

Question 1:

In 5-10 year I would like to be working at a manufacturing company and in a position where I am able to make a significant difference and take ownership over my own projects. To get to this point, I think that I will need to get experience working in smaller companies where each individual employee is required to take on more responsibility and then later decide if I want to transition to a larger company to work in larger industry roles.

I have done a lot of research on local food and beverage companies around Chicago while looking for internships. Glassdoor.com and LinkedIn have both been very useful tools to find companies that I would have otherwise never have learned about.

I do not have a prospective employer yet but would definitely be interested in pursuing a graduate degree if it were sponsored by a future employer.

Question 2:

The food and beverage industry is still adjusting to the 2011 Food Safety Modernization Act (FSMA). There are a number of different industry training courses that offer education on FSMA for industry professionals. Overall, FSMA focuses on regulating processes to prevent conditions that could lead to potential health hazards as opposed to reacting to health hazards after product contaminations have already occurred. This has lead to many manufacturers updating their safety procedures and equipment manufacturers marketing new solutions to the FSMA regulations.

In my senior lab group, some members of my group became upset because they felt that they were contributing more to the reports than other group members. They began to complain to the professor and leave other group members names off the final report. I was some what caught in the middle of the dispute and tried to help resolve it by keeping in closer contact with the underperforming group members and providing them with the data and materials that they needed to contribute more to the group. In the end, I think that this did not work out as well as it should have because everyone in the group is so busy. However, I think that this was really a failure of everyone in the group to an extent because anyone could have helped resolve the problem by taking on more leadership and organizational responsibility.

Yes, the AIChE code of ethics was discussed in the ethics section of CHE 396. The first provision of the AIChE code of ethics is that safety should always be prioritized, other provisions include acting professionally (not breaching any contracts), advising clients and employers as to the best engineering practices, accepting responsibility for your actions, and only providing professional services if you are qualified to do so.

Question 3:

In future positions, chemical engineering majors may work with electrical or civil engineering majors.

Electrical engineers may work at a chemical plant as controls engineers or be responsible for onsite electricity generation and transmission, or at a design/consulting company to help develop P&IDs or to make more detailed electrical diagrams from P&IDs. Civil engineers will likely be responsible for construction management when any new plant is built or if any project requires any major construction.

Question 4:

a.)

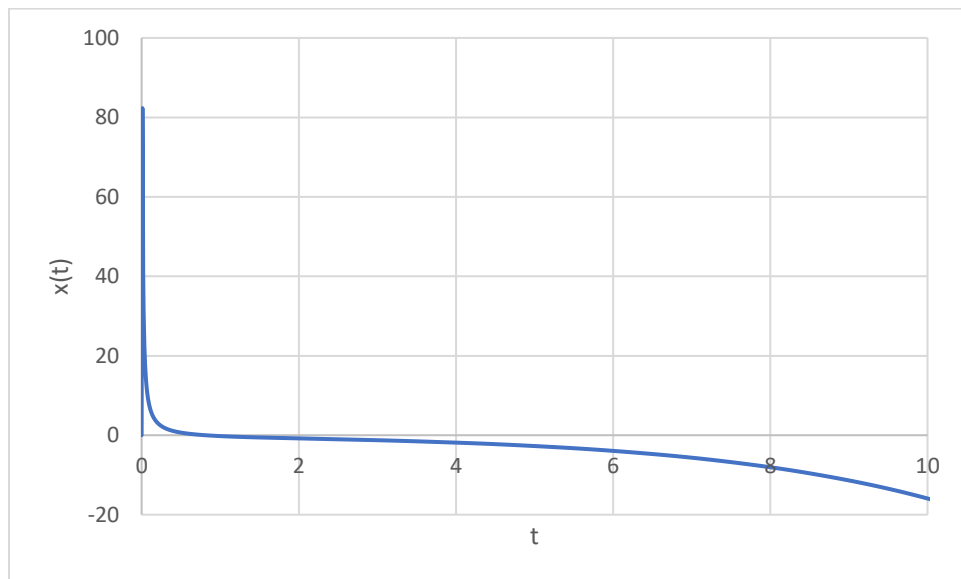
$$\frac{dx}{dt} = \frac{1}{\tau} [x_{in}(t) - x(t)] - kx(t)^2$$

$$\frac{dx}{dt} = \frac{1}{\tau} + \frac{1}{\tau} \varepsilon \sin(\omega t) - \frac{1}{\tau} x(t) - kx(t)^2$$

$$x(t) = \int \frac{1}{\tau} + \int \frac{1}{\tau} \varepsilon \sin(\omega t) - \int \frac{1}{\tau} x(t) - \int kx(t)^2$$

$$x(t) = \frac{t}{\tau} - \frac{\varepsilon}{\tau \omega} \cos(\omega t) - \exp\left(\frac{t}{\tau}\right) + \frac{1}{kt}$$

$$x(t) = \frac{t}{3.4} - \frac{0.1}{3.4(5)} \cos(5t) - \exp\left(\frac{t}{3.4}\right) + \frac{1}{1.2 t}$$



b.)

I don't think this integral has an analytical solution, but I was able to solve it in WolframAlpha as

$$\int_1^7 \frac{dx}{1 + \cos^3(x) + \ln(x)} = 3.1557$$

Question 5:

The mechanical energy of the water wheel made from the drop in potential energy of the water as it moves from the top to the bottom of the wheel. And so, the power output should be calculated as the drop in potential energy multiplied by the water flow rate.

Assuming that the water wheel has a radius of 4 ft based on the man's height, then it has a diameter of 8 ft (~2.5 m). Looking at the close up images, I assumed that each tray of the wheel holds approximately 1 L (0.001 m³) and that there are about 20 trays total on the wheel.

Based on these guesses, I calculated the flow rate of the water as

$$\frac{1 \text{ rotation}}{15 \text{ sec}} \left(\frac{20 \text{ trays}}{\text{rotation}} \right) \left(\frac{0.001 \text{ m}^3}{\text{tray}} \right) = \frac{1}{750} \frac{\text{m}^3}{\text{sec}}$$

From this flow rate, the power is calculated as

$$Power = P\dot{E} = \dot{m}g\Delta h$$

$$Power = \frac{1}{750} \frac{\text{m}^3}{\text{sec}} \left(997 \frac{\text{kg}}{\text{m}^3} \right) \left(9.81 \frac{\text{m}}{\text{s}^2} \right) (2.5 \text{ m})$$

$$Power = 32.6 \text{ W} = 0.0437 \text{ hp}$$

This estimate would not include energy lost to inefficiencies like frictional energy loss.